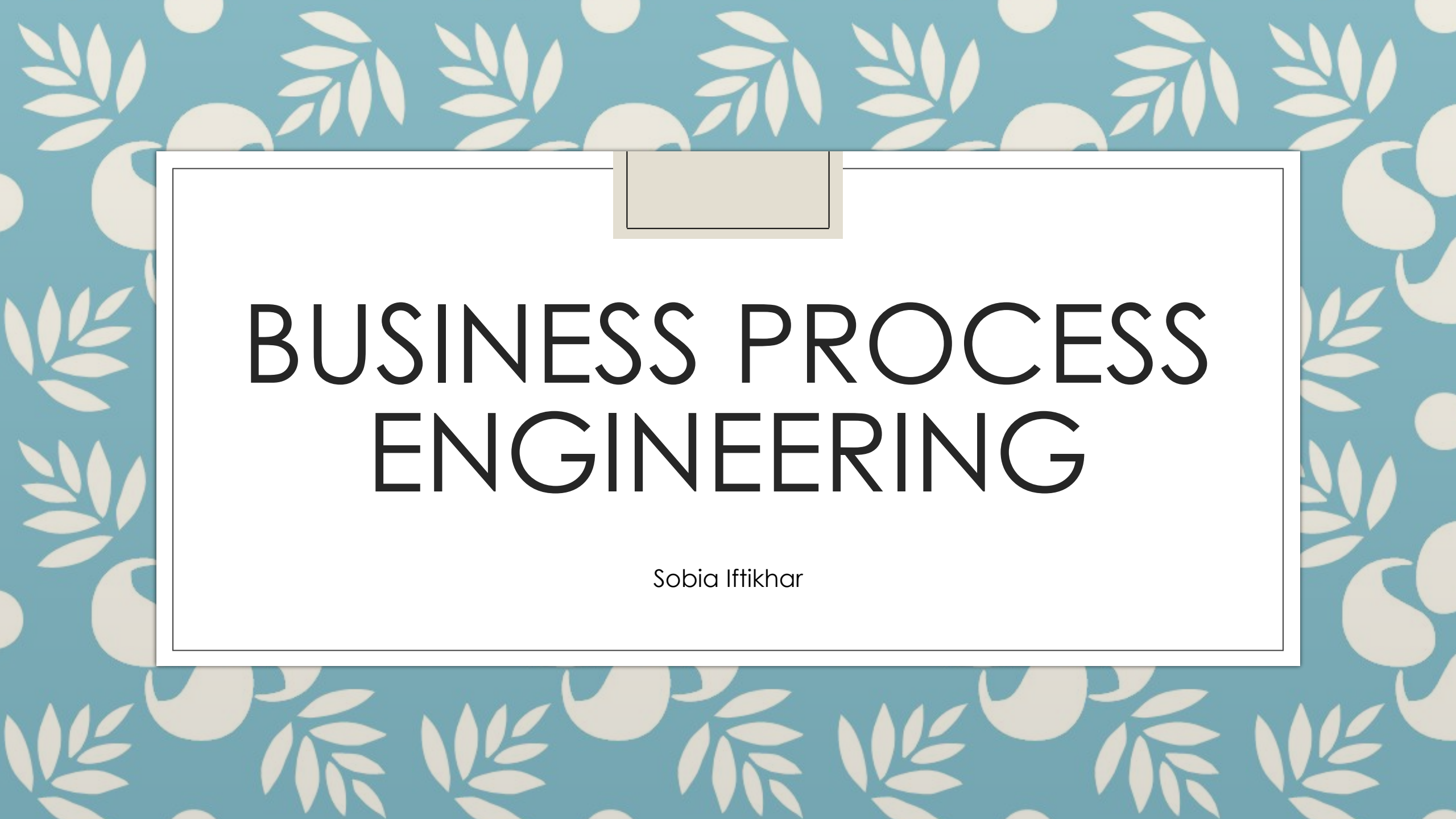




BUSINESS PROCESS ENGINEERING

Sobia Iftikhar

Content

- Properties of Business Processes & Business Process Management Architectures

Central Idea

1. The investigation of properties of business process models is an important aspect of business process management.
2. If a certain property at the business process model level can be shown, then all process instances based on that business process model expose this property.

Properties of Business Processes

- Data Dependencies
- Structural Soundness
- Relaxed Soundness
- Weak Soundness
- Lazy Soundness
- Soundness
 - Motivation of Soundness
 - Definition
 - Soundness Theorem

Data Dependencies

- Application data are an integral part of business processes.
- Data can be created, modified, and deleted during the execution of business processes.
- Since business processes consist of a set of activities that are related, these activities operate on an integrated set of application data.

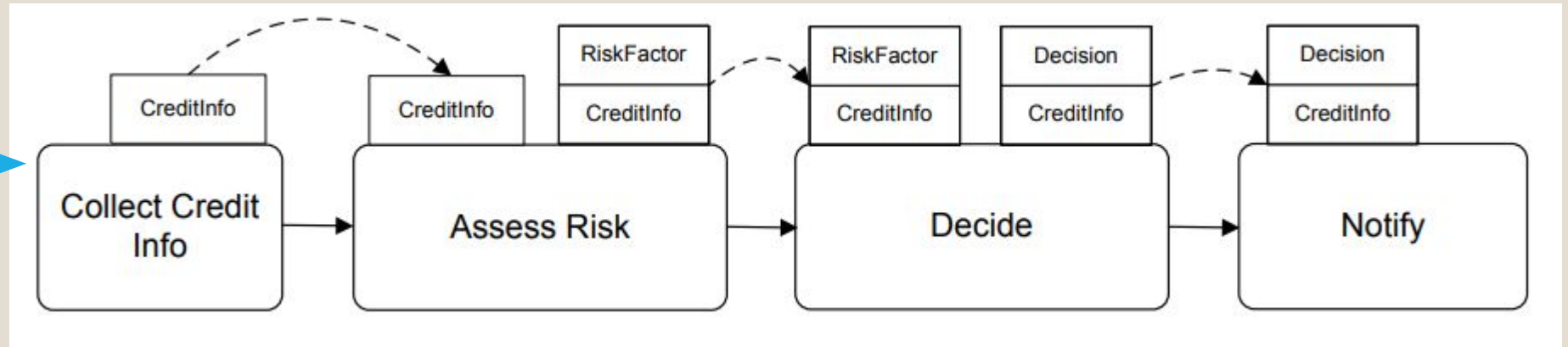
Data in business process models has two aspects, both of which need to be covered:

- Data that activity instances manipulate by invoking applications or services.
- Data dependencies between process activities.

Data Dependencies

There are data dependencies between the activities mentioned

. Only when
this data is
available

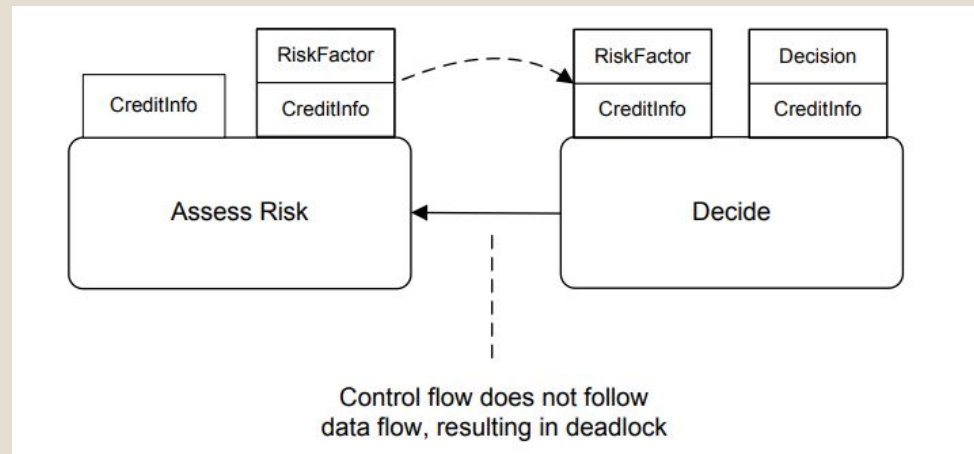


There are data dependencies between the activities mentioned. The Collect Credit Info activity is the first activity performed. Only when this data is available, can the risk be assessed in the Assess Risk activity, the final decision be made (Decide), and the requestor be notified (Notify). Therefore, the ordering of the activities in the business process is strongly related to the data dependencies of the activities.

An example of data flow in a business process in the financial sector is given. A credit approval business process contains activities to enter a credit request, to assess the risks of granting the credit, and to inform the customer about the decision made by the financial institution.

Data Dependencies

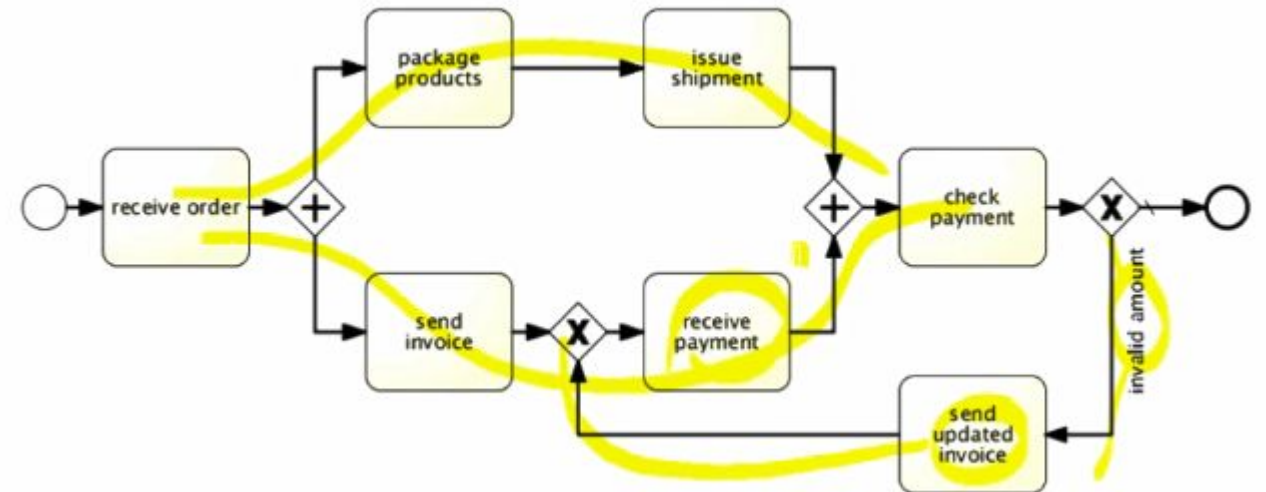
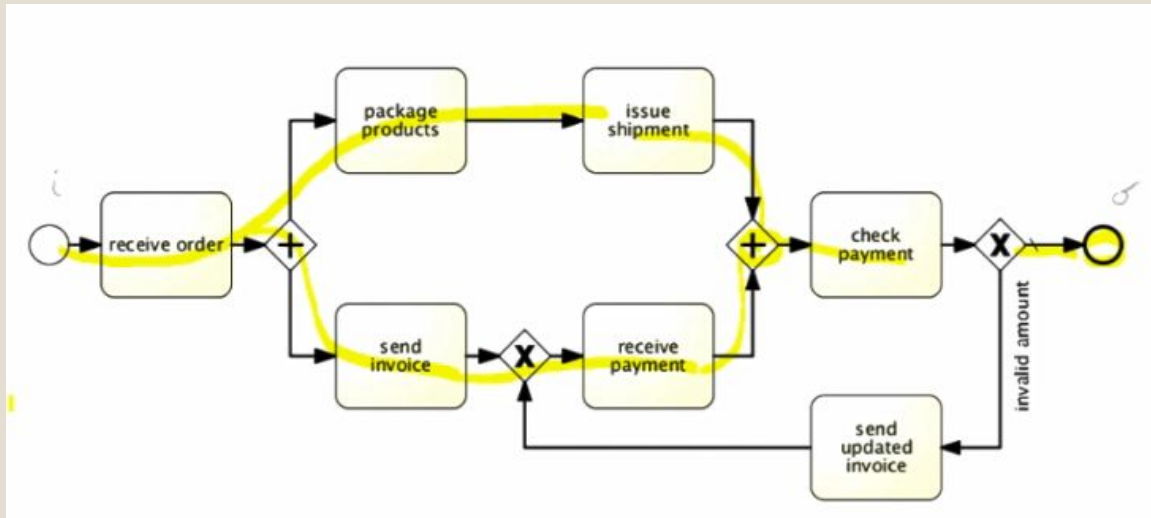
- This property is known as control flow follows data flow.
- Control flow needs to follow data flow, since otherwise the process instance would come to halt.
- neither of these activities can be started, because control flow defines that Assess Risk can only start after Decide has completed, and Decide can only start after Assess Risk has generated the risk factor data value.



Unsound Behaviour

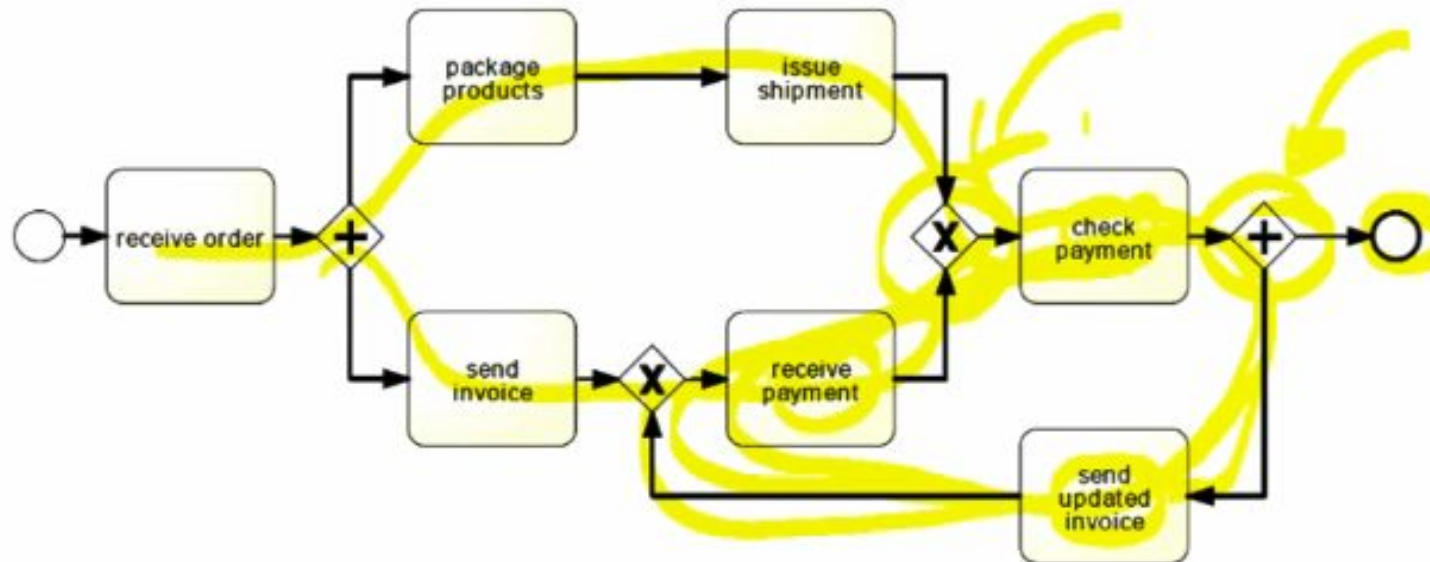
- We can detect unsound process behavior by finding a violation of either of the soundness properties
- Violation of (1)
 - The process cannot reach the final event o , it gets stuck before
 - Deadlock (before o is reached)
- Violation of (2)
 - Remaining activity in the process after reaching o
 - Or even infinite activity, livelock
- Violation of (3)
 - Activities that can never be executed
 - Dead activities

Violation 01-deadlock

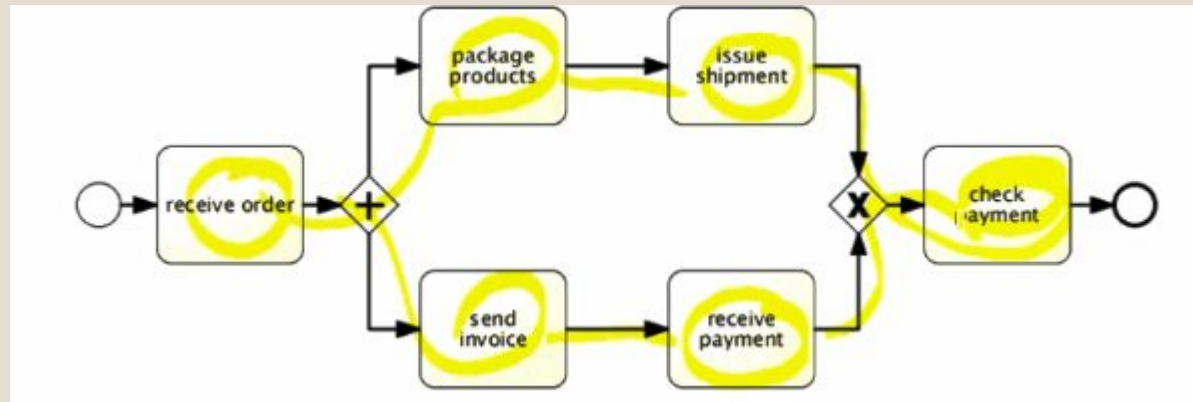


Violation 02

- A process is in a livelock, if the control flow structure makes the process executing an infinite number of activities



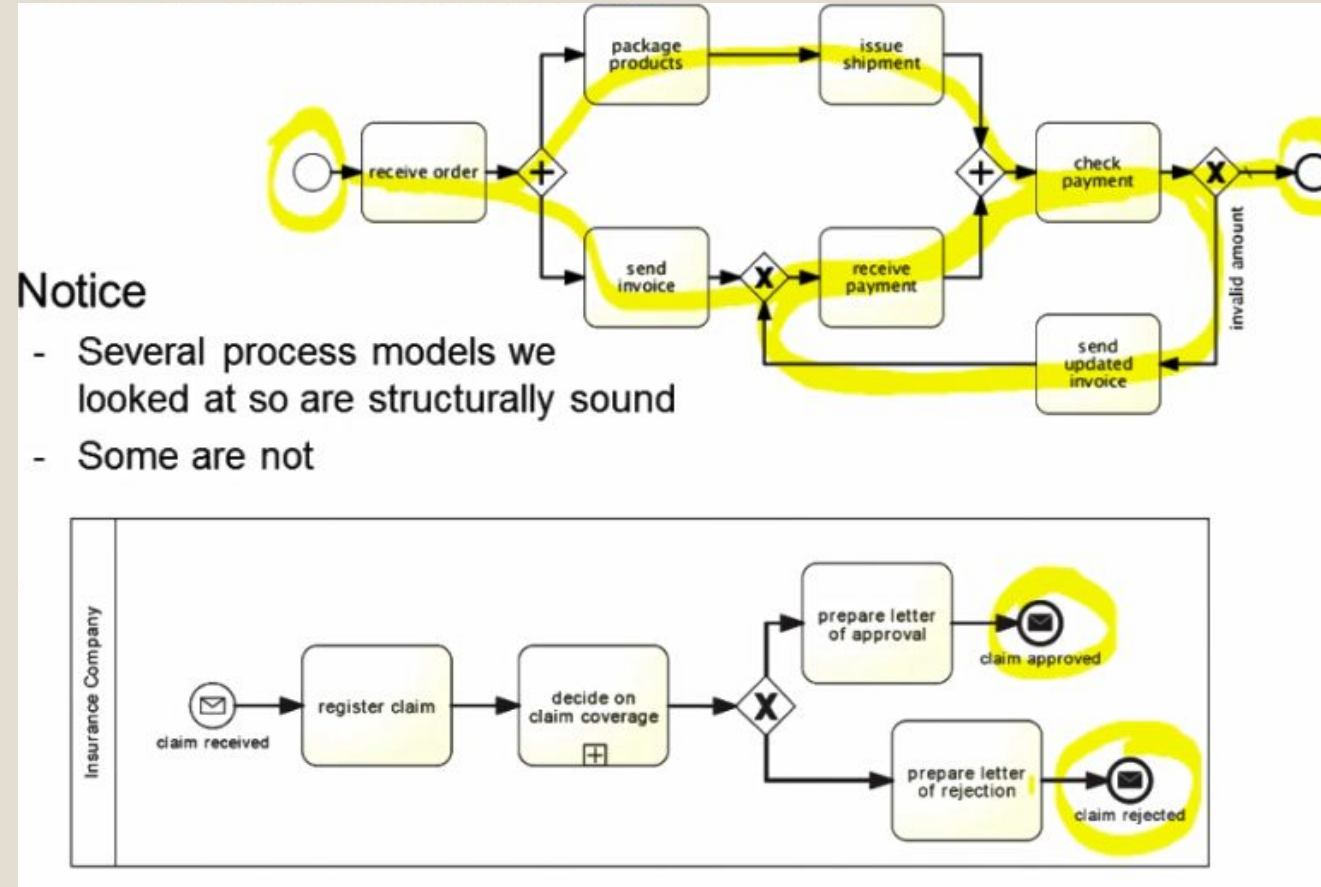
Violation 03-dead activities



Structural Soundness

- Soundness assume that a process is structurally sound.
- A process model is structurally sound if the criterion that a process model should fulfill. following conditions hold:
 - There is exactly one initial node, which is only node without any incoming edges. (i)
 - There is exactly one final node, which is only node without any outgoing edges. (o)
 - Each node in the process model is on a path from the initial node to the final node. (i) to (o)

Example

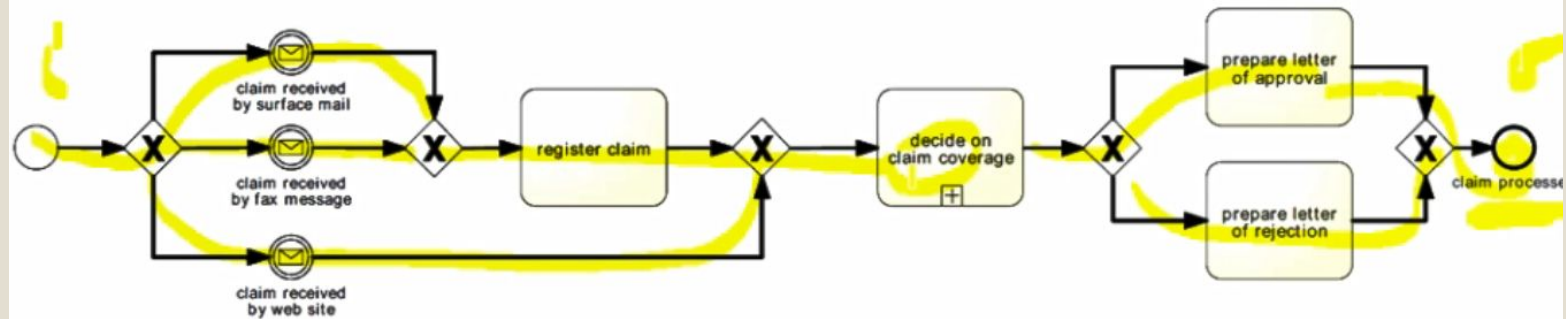


Soundness criterion

- We assume process models to be structurally sound
- Soundness
 - (1) Once a process has started, regardless of which decisions are taken by the process, at some point, the process will reach the end event o
 - (2) If and when it reaches o , the process has completely terminated, i.e., no further activities can be executed by the process anymore
 - (3) Each process activity participates in at least one execution (in one trace)
- Notice
 - Soundness is a generic correctness criterion, similar to normal forms in databases
 - It makes sure that all process instances behave correctly

Soundness criterion-Properties

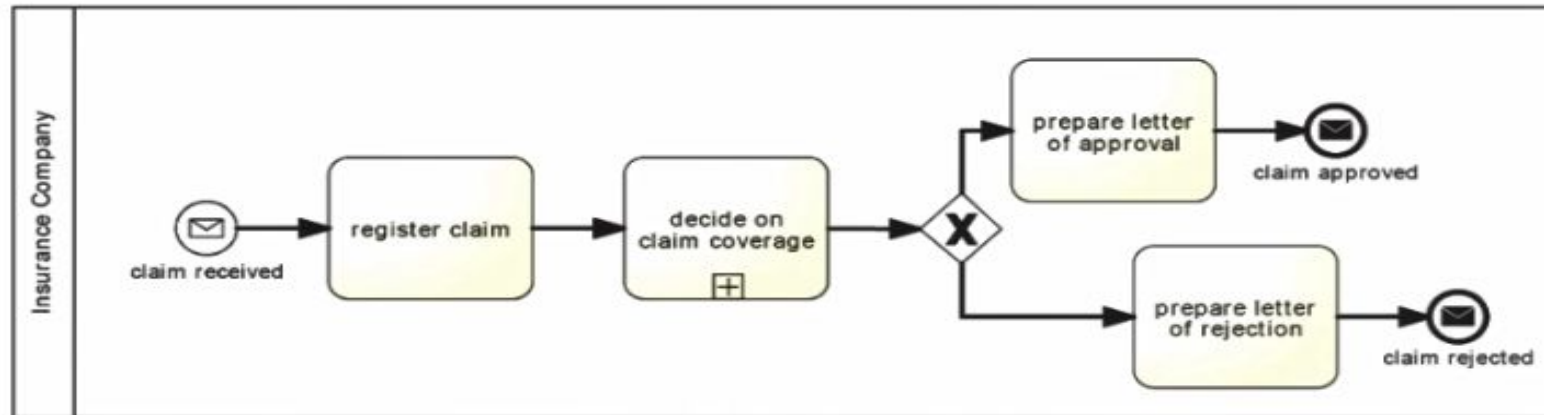
- Properties (1), (2), and (3) by example



- Formally, the process is in a deadlock after o has been reached
 - But this behavior is intended, so this does not violate soundness!

Structural Soundness

Process models with multiple end events are not structurally sound

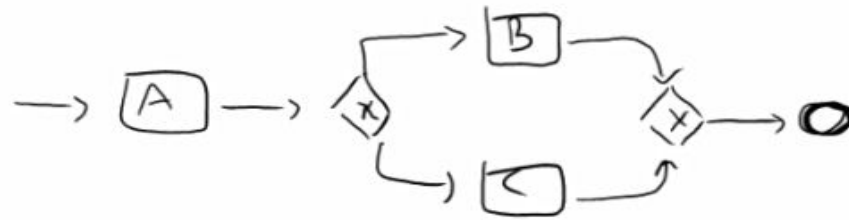


All these process models cannot be not sound

- Even if they do not exhibit any behavioral anomalies
- Therefore, we transform these models to structurally sound process models without changing their behavior

To start formal analysis

- Now we transform the structurally unsound process into structurally sound process.

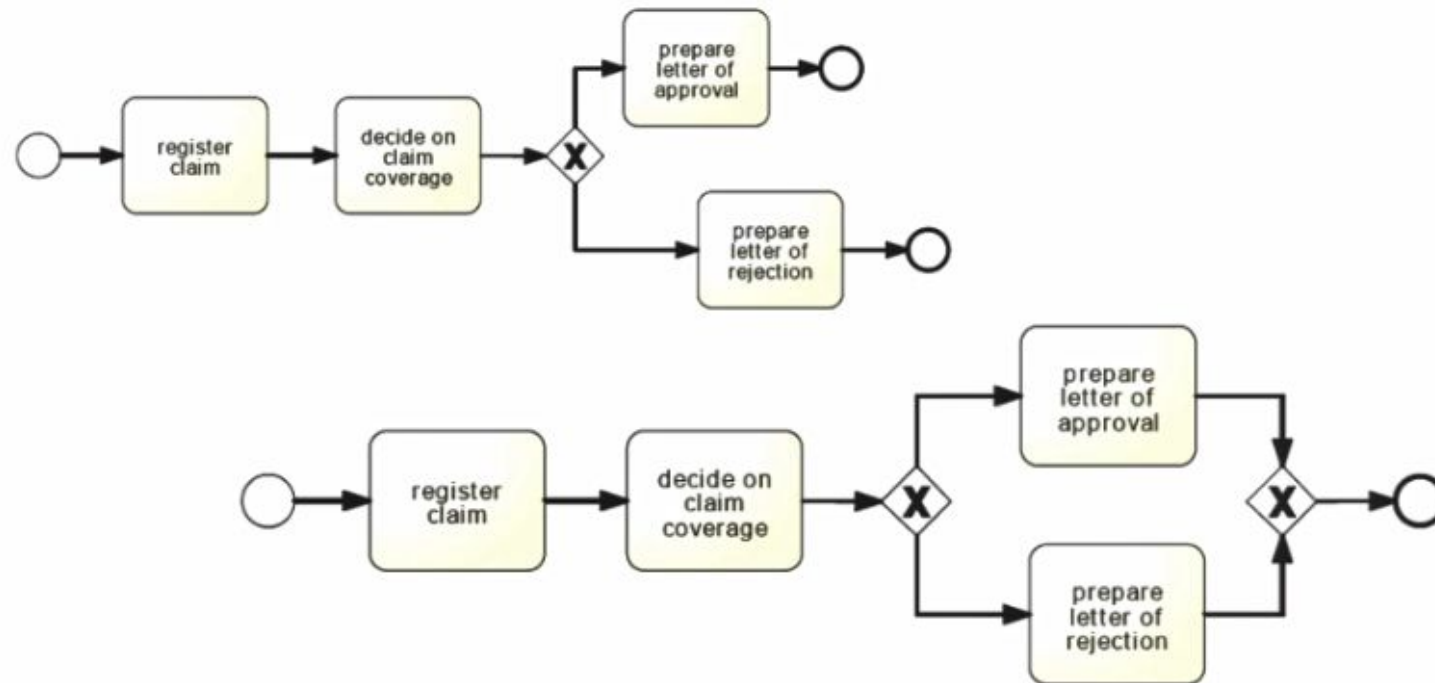


- Process models with exclusive end events can be translated to structurally sound process models by
 - deleting the end events,
 - introducing an exclusive gateway that joins the end event branches, and
 - introducing a new end event.

Structural sound

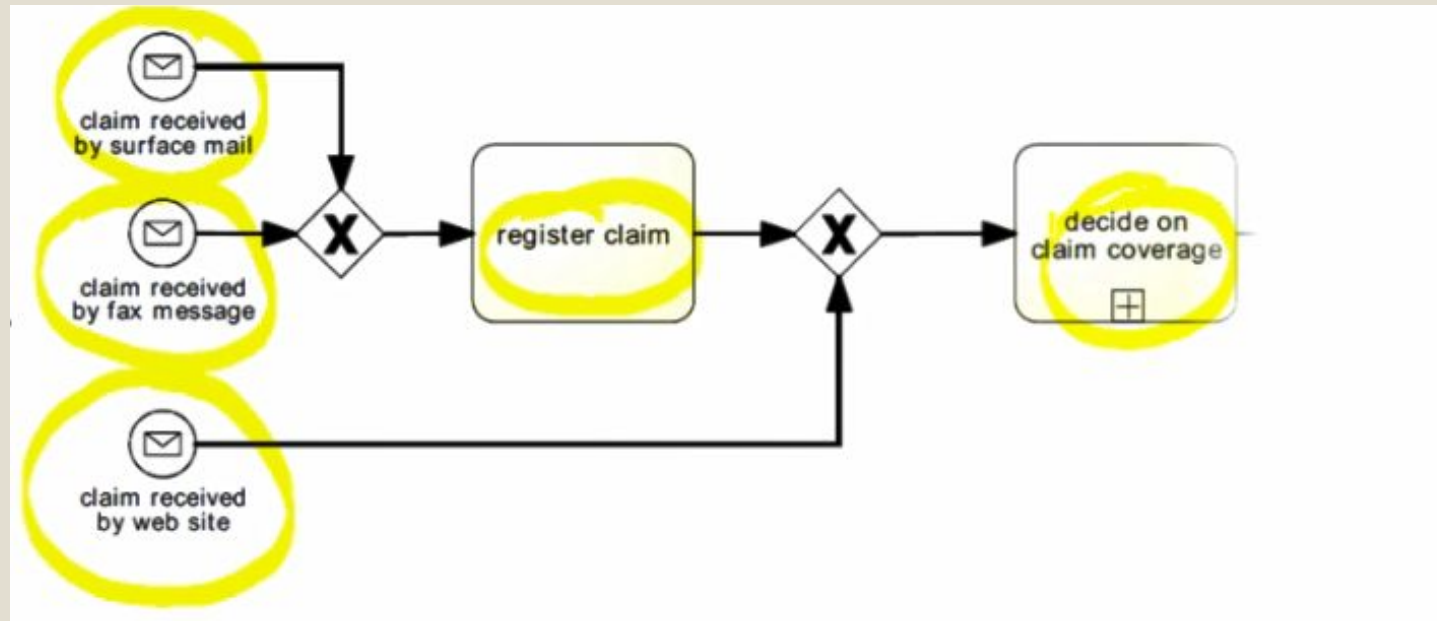
- Notice

- The different outcomes of the process are no longer represented explicitly in the process model
- Since we use the transformed model only for analysis, this is fine



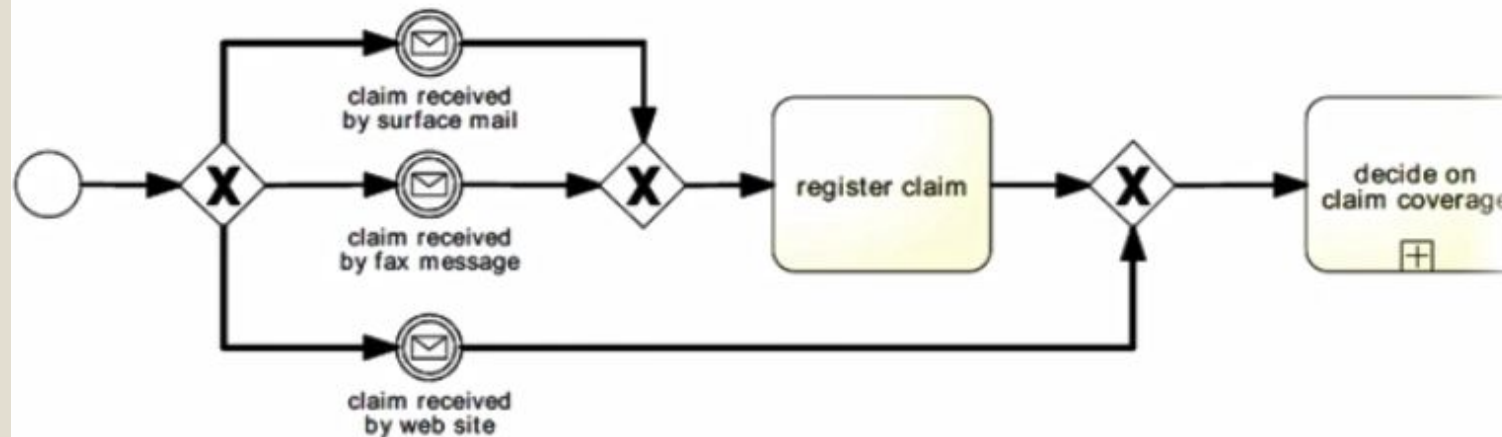
Multiple start Events.

- We can see process models with multiple start week.

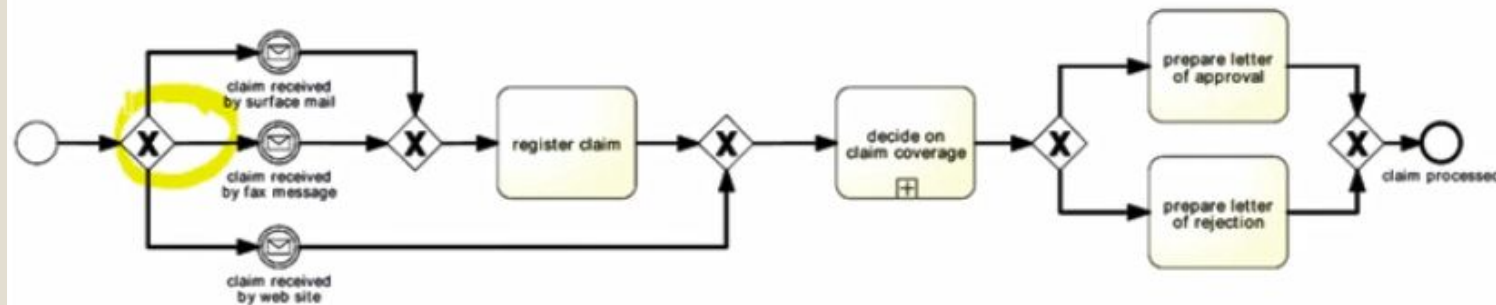


How to transform

- Also in this case, we can transform processes with alternative start events to be structurally sound
 - Add a new start event and an exclusive gateway
 - Connect it to all alternative start events, which become thereby intermediate events



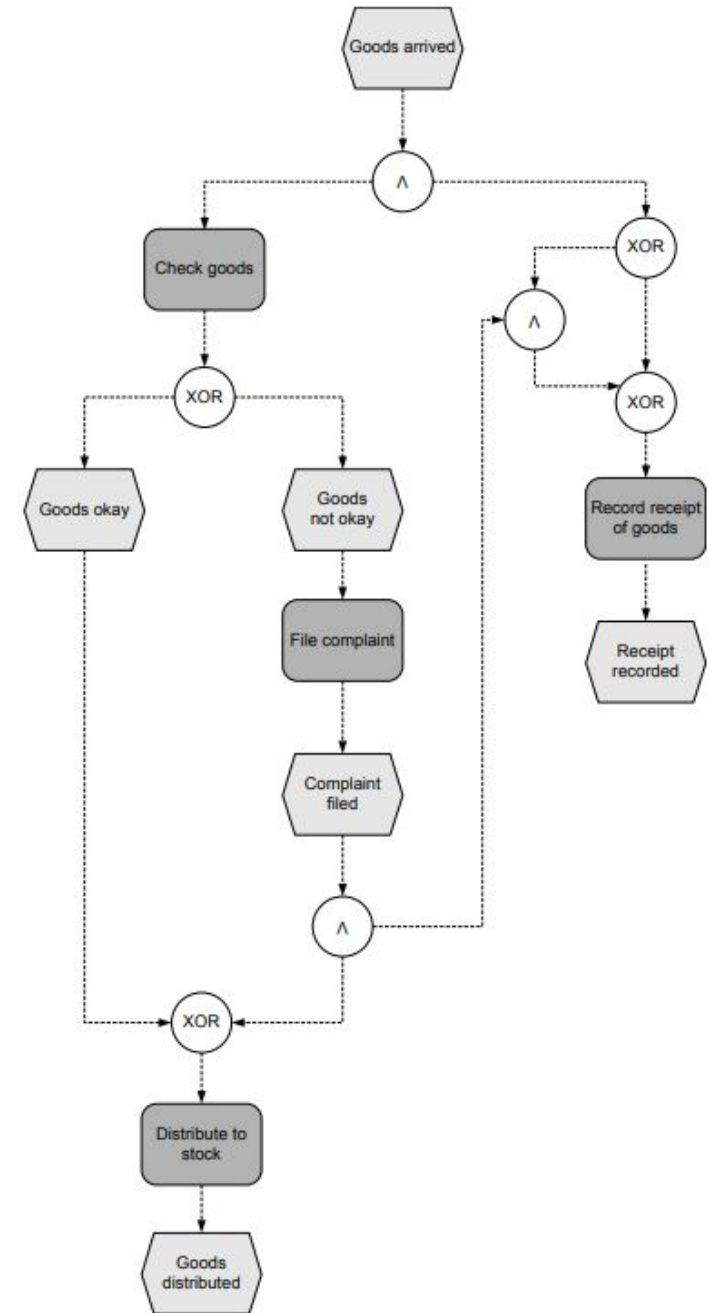
- These transformation operations lead to a process model that is structurally sound



Relaxed Soundness

- The idea of relaxed soundness is that for each transition there is a firing sequence that relaxed soundness allows business process models that expose deadlock behavior
- A process is relaxed sound if each task of the business process is part of a properly terminating sequence..

- If the exclusive or split selects the and join branch but the goods are okay, the and join waits permanently for the filed complaint, which will never appear. As a result, the process is stuck in a deadlock situation.
- If the exclusive or split does not select the and join branch, but the goods are not okay, the and split cannot fire, because just one incoming edge—the filed complaint—is signalled. In this case, the receipt of the goods is registered, but the complaint on the quality of the goods is lost

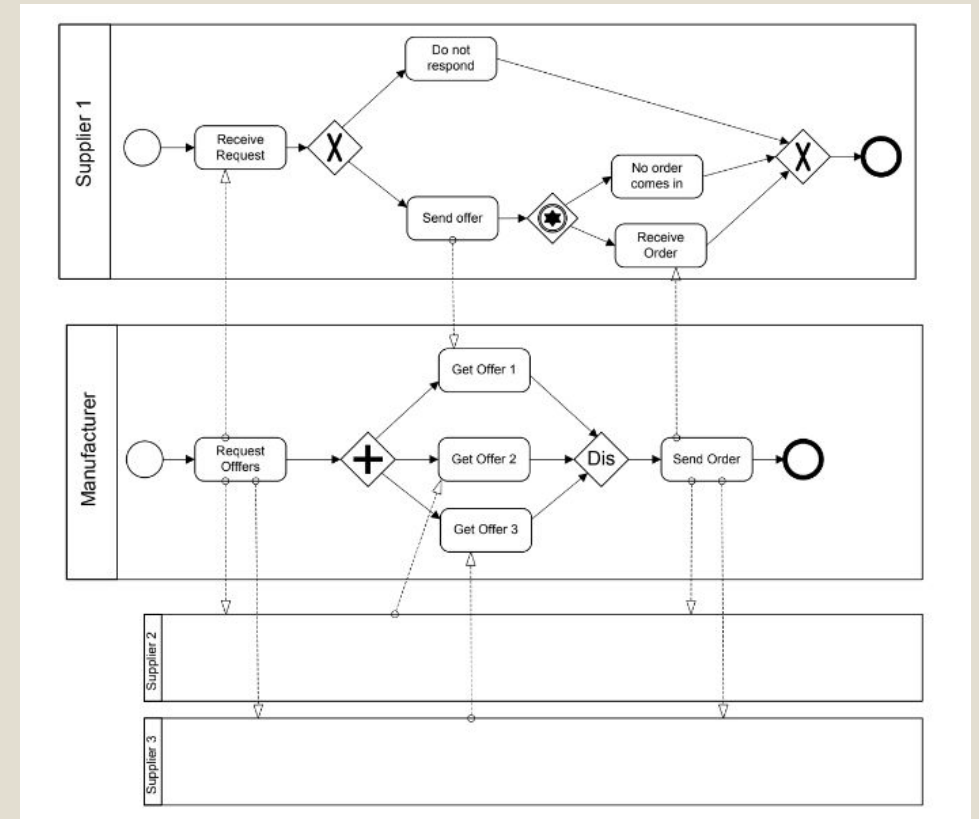


Weak Soundness

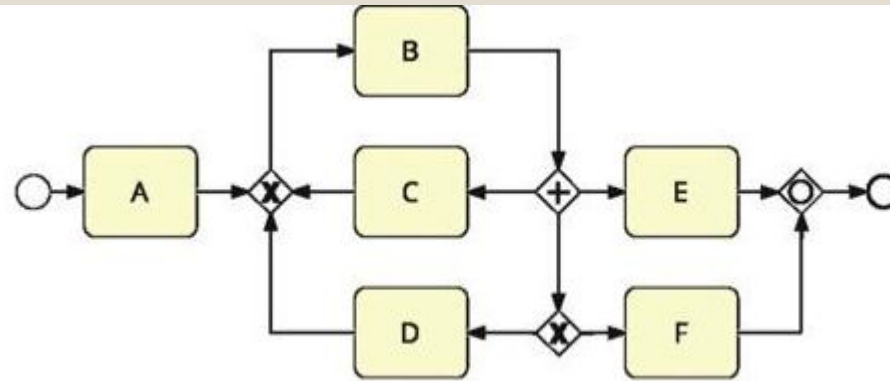
- Weak soundness disallows deadlocks, but it allows certain parts of the process not to participate in any process instance,
- A workflow system (P, N, i) is weak sound if and only if the following holds:
 - • For every state M reachable from state i there exists a firing sequence leading from M to o , i.e.,

Lazy Soundness

- assume that the offers received from Suppliers 1 and 2 have led to the activation of the join. Further assume that the offers from Suppliers 3 and 4 arrive only after the final event has been reached. Then, receiving the offers for Suppliers 3 and 4 and paying them are lazy activities.

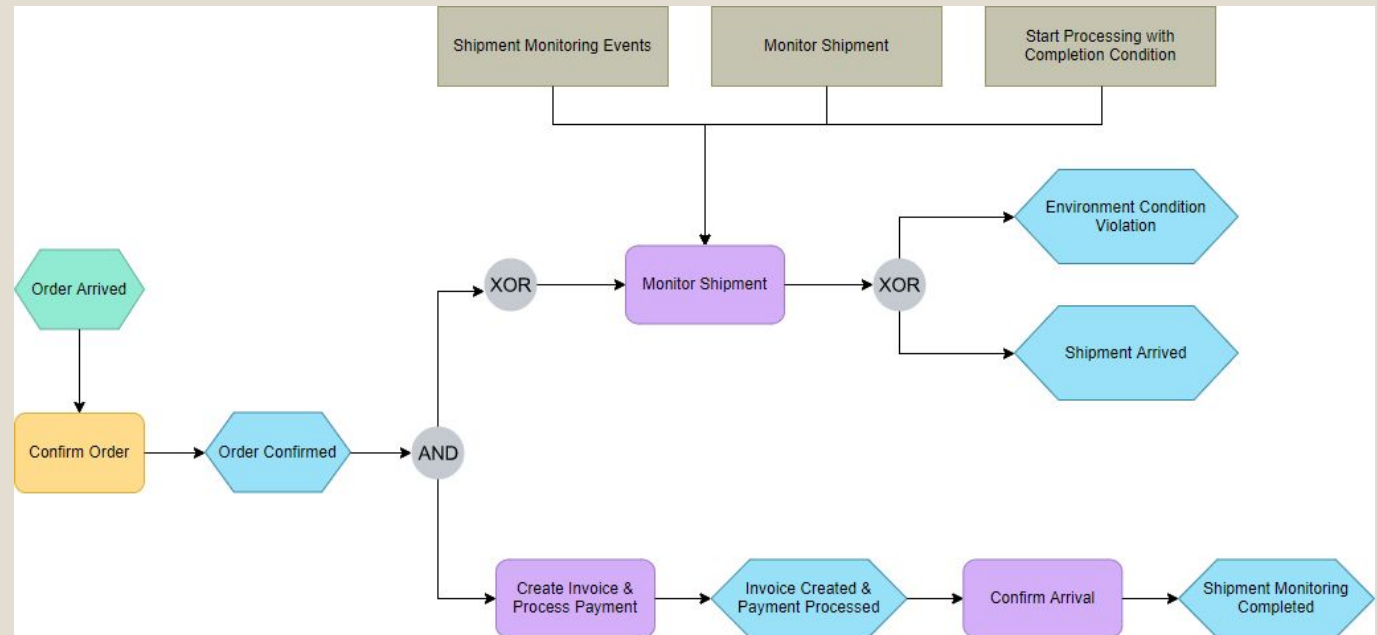


Exercise



Process Notation

- BPMN Notation
- Event driven process chain (EPCS)
- Petri nets

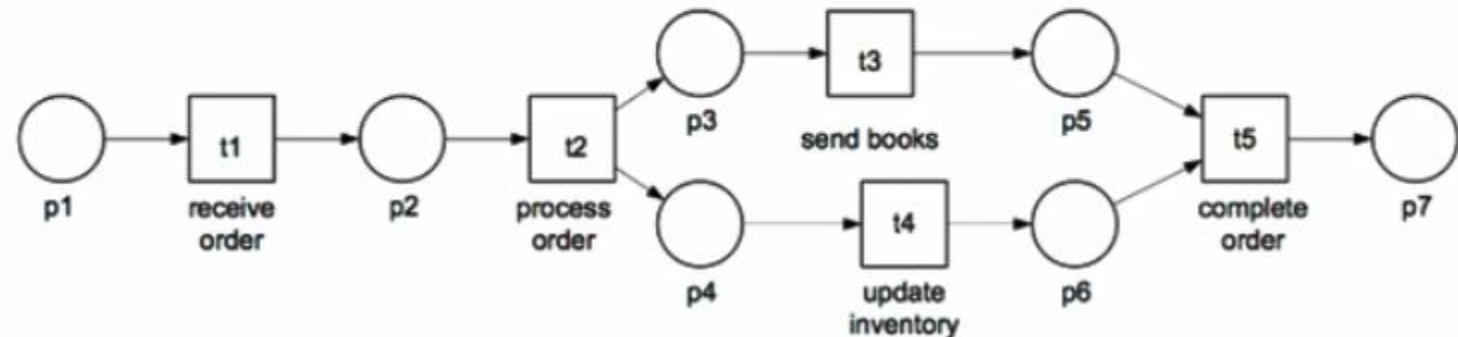


Petri nets

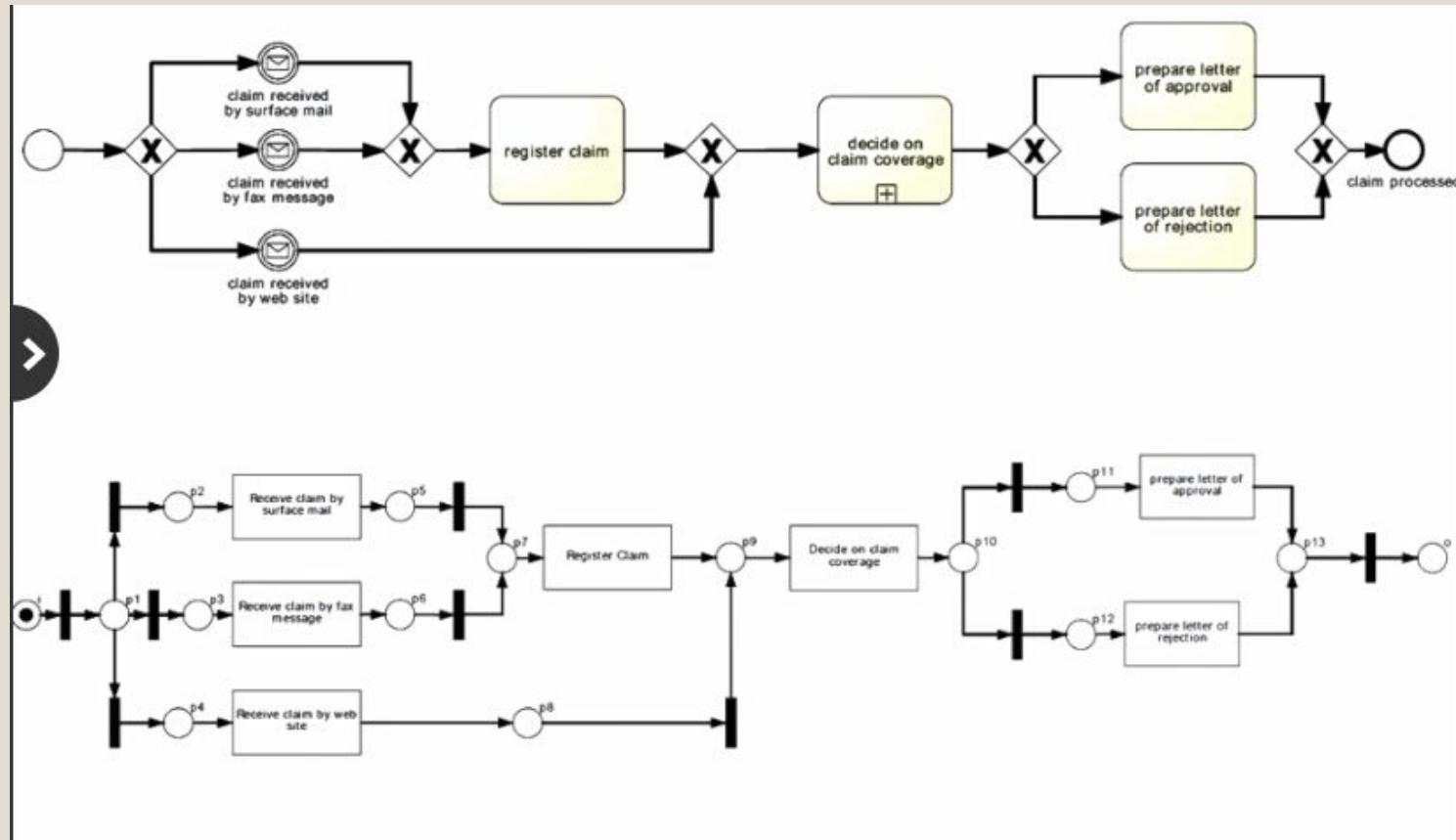
- Petri nets are directed graphs devised to represent the concurrent behavior of dynamic systems.
- The behavior of Petri nets is represented by tokens moving over the places of petrinet
- Should be free of obvious anomalies of soundness.

- **Terminology**

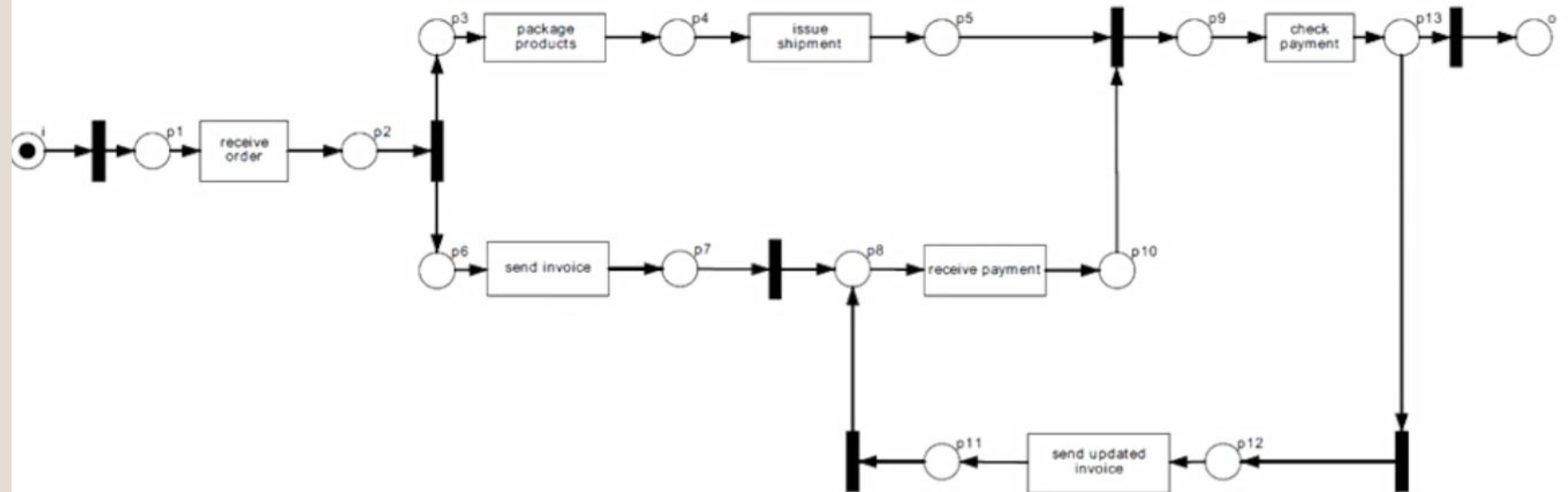
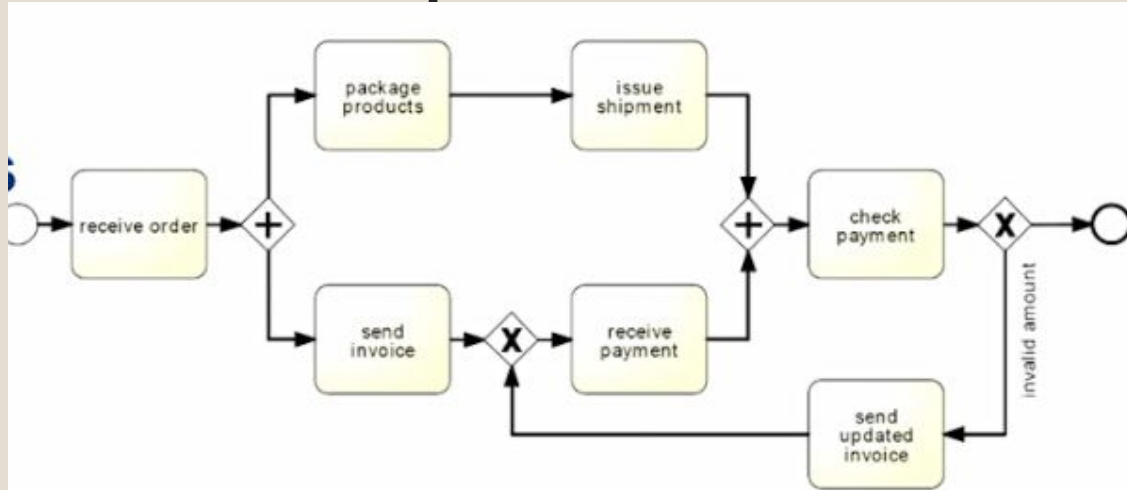
- p1 is input place of t1
- p3 and p4 are output places of t2



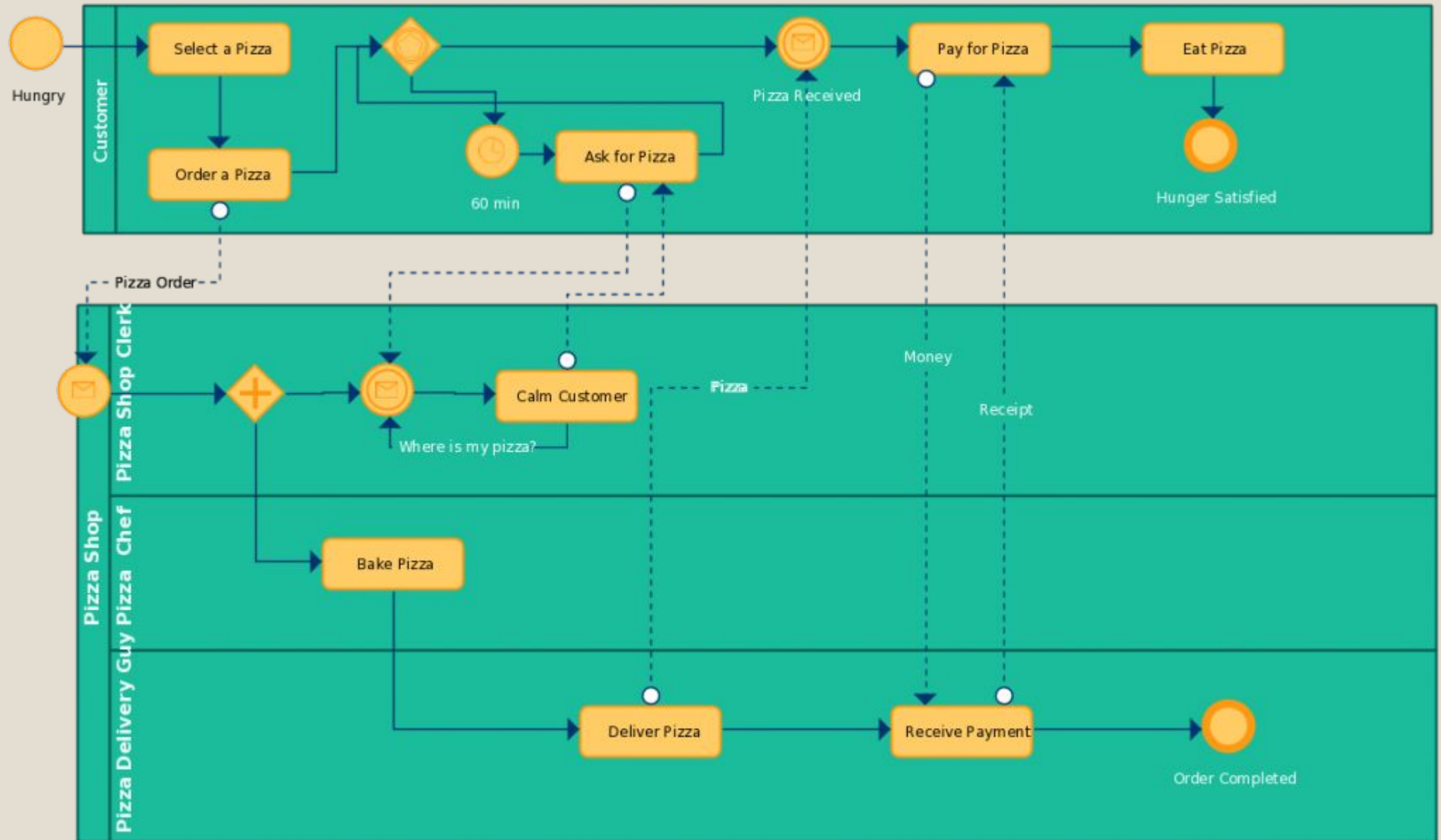
Petrinets : Modeling Representation



Example



Example



<https://app.creately.com/diagram/TdR0Mb7tj1t/edit>



 Folders

 Templates

 Shapes

 Images

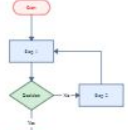
 Google Search

 Icons

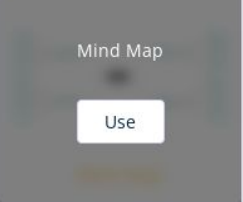
Welcome sobia!



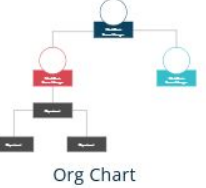
Featured Templates



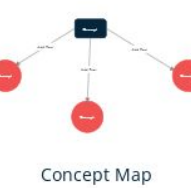
Flowchart



Mind Map



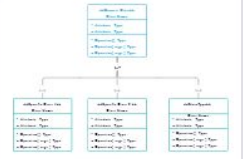
Org Chart



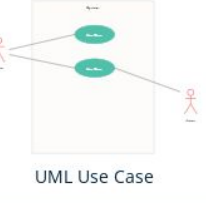
Concept Map



Infographic



UML Class



UML Use Case



Database

Templates

Popular Tools

Brainstorm Ideas

Improve Plans & Strategies

Visualize Projects & Tasks

Design Software and Systems

Visualize IT Infrastructure

Organize People & Processes

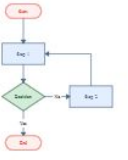
Run Meetings & Workshops

Document Research

Grow Sales Productivity

Tools for Educators

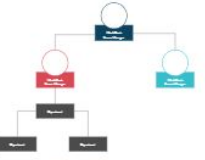
Insert a Template



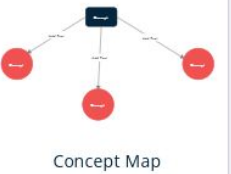
Flowchart



Mind Map



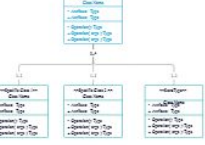
Org Chart



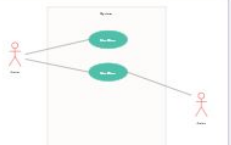
Concept Map



Infographic



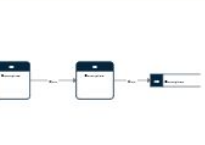
UML Class



UML Use Case



Database



Dataflow Diagram



Network Diagram



AWS Architecture



GCP Diagram

Universal Process Notation

